

Optimal Unemployment Insurance, with Human Capital Depreciation, and Duration Dependence.

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April 2004

Abstract

This paper studies the effect of human capital depreciation and duration dependence on the design of an optimal unemployment insurance (UI) scheme. Our results partially confirm those obtained in most previous studies: benefits should decrease with unemployment duration. The optimal program also generates two main novel features, which are not present in stationary models. First, if human capital depreciates rapidly enough during unemployment, UI transfers are bounded below by a minimal “assistance” level that arises endogenously in the efficient program. Second, we study the optimality of imposing a history contingent wage tax after reemployment. Our numerical simulations based on the Spanish and US economies suggest that the wage tax should decrease with the length of worker’s unemployment spell, and become a *wage subsidy* for long-term unemployed workers. As a by-product of our analysis, we develop a systematic approach suitable for studying the properties of the value function associated with a wide range of dynamic moral-hazard problems, and other models with similar characteristics.

Keywords: Unemployment insurance, human capital depreciation, duration dependence, recursive contracts, moral hazard, envelope theorems.

JEL Classification: C61, D63, D82, D83, J24, J31, J38, J64, J65.

*University College London and Institute of Fiscal Studies. I would like to thank Antonio Cabrales, Per Krusell, Matthias Messner, Kjetil Storesletten and Fabrizio Zilibotti for many helpful comments. The paper benefits from a profitable discussion with Philippe Aghion and Gianluca Violante, during the preliminary stages of the work. I wish to thank conference and seminar participants at the Duke University, the LSE, the Northwestern University, the University of Iowa, the Universitat Autònoma and the Universitat Pompeu Fabra in Barcelona, the IIES in Stockholm, the CERGE-EI in Prague, the IZA institute in Bonn, and at the UCL. All remaining errors are mine. This research was supported by Marie Curie Fellowship MCFI-2000-00689, and the Spanish Ministry of Science and Technology Grant BEC2001-1653. *Correspondence:* Nicola Pavoni, Department of Economics, UCL, Gower Street, London WC1E 6BT. *E-mail:* n.pavoni@ucl.ac.uk

1 Introduction

Unemployment insurance (UI) programs are an important ingredient of social welfare policies in developed economies. Public expenditures devoted to UI and social assistance programs in the OECD countries exceeded 2% of GDP in the mid 1990s (Martin, 2002). These programs have been widely criticized because of the adverse effects they can have on worker's incentives to search for a new job. This criticism has stimulated extensive research into optimal insurance schemes that take these perverse effects into account.

A series of papers use the dynamic moral hazard model to analyze the trade-off between (unemployment) insurance and (search) incentives. In their seminal work on UI, Shavell and Weiss (1979) establish that, because of moral hazard, benefits must decrease throughout the unemployment spell and approach zero in the limit. Hopenhayn and Nicolini (1997a,b) extend the analysis of Shavell and Weiss by introducing the possibility of contingent wage taxes after reemployment and confirm the decreasing benefits result. In addition, the analysis of Hopenhayn and Nicolini suggests that up to a 30% of overall spending in unemployment compensations could be saved by introducing a tax on the wage the agent receives after reemployment that *increases* with the length of the previous unemployment spell. In Pavoni (2003), I notice that the optimal schemes based on dynamic moral hazard such as the one proposed by Hopenhayn and Nicolini implicitly assumes that the planner can (and will) inflict infinite punishments on workers.¹ To design an implementable UI scheme, I consider the possibility that the planner must respect an exogenously given lower bound on the expected discounted utility that the agent can have ex-post, regardless of the previous history. Among other properties, I find that the UI transfers should be bounded away from zero and that the optimal wage tax is roughly constant during unemployment.

In all these models, the key features of the environment are that the probability of finding a new job depends *only* on the (unobservable) search effort exerted by the agent, and that the available gross wages distribution is constant throughout unemployment spells. It is well documented that job opportunities deteriorate during unemployment.² In fact, many

¹The key properties of the standard stationary model are reproduced in Proposition 1.

²We will extensively revise the literature on wage depreciation, job displacement and duration dependence below in this section. However, some of our calibrations are based on Keane and Wolpin (1997) who use NLSY data and estimated, structurally, an annual human capital (wage) depreciation rate for white US males during unemployment of between 9.6% (for blue collars) and 36.5% (for white collars). In addition, many authors consistently find that displaced US workers face a large and persistent earning loss upon reemployment in the order of 10 – 25% compared with continuously employed workers (Jacobson et al., 1993; Ruhm, 1987; and Bartel and Borjas, 1981). Finally, Van den Berg and van Ours (1996), after controlling for unobservable

OECD countries propose and apply active labor market policies and wage subsidies for long term unemployed workers, mainly because of this adverse change in job opportunities.³ We thus believe that human capital depreciation and unemployment duration dependence are important elements, that need to be included in the study of an optimal UI designing problem. In this paper, we extend the basic model of unemployment insurance with moral hazard to allow for both the gross wages and the probabilities of re-employment to depend on the length of the worker’s unemployment spell. Although we do not analyze how the UI designing problem interacts with other welfare or labor market policies, we do allow the planner to impose history contingent wage taxes or pay wage subsidies upon reemployment.

Our study confirms one important result of most previous studies with stationary models: benefits should decrease with unemployment duration. In fact, this behavior characterizes virtually all existing UI schemes in OECD countries. Moreover, we find that very simple schemes, defined by a low constant benefit payment b and a higher time-invariant wage w , can never be optimal. This result holds for any reasonable range of parameters (no restrictions on the worker’s utility function, but concavity, are required), and regardless of the characteristics of the human capital depreciation process. In exchange, we propose a simple necessary characteristic of any optimal unemployment insurance program. A “back of the envelope” test of optimality.

The introduction of human capital depreciation and duration dependence also generates two main novel features in the optimal program, which are not present in stationary models. First, provided that human capital depreciates sufficiently rapidly during unemployment, the optimal path for unemployment benefit payments is initially decreasing and then becomes completely flat. The idea is that, for low levels of human capital, the planner does not find worthwhile to induce the worker to search intensively for a new job. Hence, UI benefits eventually stop decreasing and remain constant forever since the (long-term) unemployed worker is fully insured. The presence of a minimal, “assistance” level of UI benefits generates an endogenous lower bound on worker’s expected discounted utility, which provides an alternative way of eliminating the “immiserization” result of Thomas and Worrall (1990).⁴ This feature

heterogeneity, find that the US white male exit probability “genuinely” decreases by 30% after 3 months of unemployment.

³Although the programmes differ substantially, there are mainly eight OECD countries that have actually introduced major welfare-to-work programmes: the United States, Canada, the United Kingdom, Ireland, Denmark, France, the Netherlands and Sweden. In several others countries, such programmes are under careful consideration.

⁴In the literature of repeated moral hazard, the immiserization result is usually eliminated by *exogenously*

of the optimal contract creates an important link between the characteristics of the efficient UI scheme and the speed of skills depreciation in the economy, which can also be used for positive analysis.

Second, recall that in our model the planner can impose history-contingent wage taxes upon re-employment. The results of our numerical exercises for the Spanish and the US economies show that in an optimal UI scheme the level of wage taxes decreases with the length of worker's previous unemployment spell and becomes a *wage subsidy* for long-term unemployed workers. The optimality of a wage subsidy is a key normative result of our non stationary search model with moral hazard. Notice that this policy advice does not come from equity considerations, nor is induced by the presence of "institutional" labor market imperfections such as the minimum wage. In our model, the worker is paid according to his productivity, and the optimality of a wage subsidy only arises from efficiency considerations. The planner finds profitable to induce the agent to search actively for a job, which will then be subsidized, since the wage subsidy cost is considerably lower than that of paying the UI benefits.

In our non stationary model at least three new forces are at work that contrast the tendency toward the wage tax increase, key property of the stationary model. First, since the planner tends to insure agents against wage depreciation, a reduction in the gross wage received by the worker decreases the optimal wage tax. Second, in our model, a reduction (depreciation) in human capital reduces the effectiveness of the search activity increasing the incentive costs. This latter effect tends to widen the difference between unemployment benefits and *net* wage, decreasing both the UI transfers and the wage tax upon re-employment. These characteristics of an optimal scheme are reinforced by a third effect, which is strictly related to the exogenous minimum bound analysis of Pavoni (2003). The (now endogenous) presence of lower bound on worker's expected discounted utility shorthands the effective time-horizon of the problem. This reduces the possibility of giving dynamic incentives, and forces the planner to design a scheme biased toward the "static" component of the incentives. The planner is hence induced to increase the within-period gap between the unemployment insurance benefit and the net wage, further reducing the wage tax.

Finally, we would like to mention that the problem analyzed in this paper required to

imposing minimum bounds on expected discounted utility the worker can have ex-post, in the optimal scheme (see, for example, Pavoni, 2003; Atkeson and Lucas, 1995; Phelan, 1995). Other authors use overlapping generation models (see, for example Phelan 1994).

solve recursively, in a possibly non-stationary framework, the dynamic hidden-action moral hazard problem. We find that the value function associated to this recursive problem is, in general, non-concave and non-differentiable. In spite of these non-smoothness problems, we show that the optimal contract can be characterized using the usual first order conditions. To derive the properties of the associated value function we develop a new approach that uses the generalized envelope theorem of Daskin (1967), and which can be extended both to an even more general class of moral-hazard problems and to other models with similar characteristics.

Literature on recursive contracts Our methodology follows the recursive contract literature, and the references most related to our approach are: Spear and Srivastava (1988), Fudenberg et al. (1990), and Phelan and Townsend (1991). In this paper, we formally study the shape of the value function associated with the dynamic moral hazard model. In their seminal work, Spear and Srivastava (1988) discuss conditions under which the value function of the dynamic model with a continuum of outcome realizations, is concave. Our approach permits to study the case with a finite number of output realizations, which is an inherent characteristic of the unemployment insurance designing problem studied in this paper. Our non-concavity result contrasts with that of Spear and Srivastava. Finally, Phelan and Townsend (1991) allow for lotteries over effort and payments, which imply the (weak) concavity of the value function. As in most applied studies, we focus on the deterministic payments case.

Literature on optimal unemployment insurance The literature on optimal UI is relatively new, yet quite extended. However, most of the papers address questions and/or use approaches that cannot be directly related to our own. The interested reader can refer to the interesting summary of Karni (1999). Atkeson and Lucas (1995) use a recursive approach to characterize the optimal contract in a stationary pure adverse-selection setup with temporary (one-period) job offers. They are mainly interested in income distribution, and their approach is closely related to that in Hansen and Imrohoroglu (1992) and Wang and Williamson (2002), where the goal is to quantify the welfare effects of unemployment insurance in general equilibrium. Wang and Williamson (1996) provide calibrated repeated OLG model with moral hazard associated with search effort and job retention. They assume that each new labor-force entrant obtains a prespecified level of ex ante utility and obtain a non monotone unemployment benefits behavior. Hopenhayn and Nicolini (2002) analyze the effects of worker's employment history on the optimal design of unemployment insurance contracts, and Zhao (2001) intro-

duces moral hazard also associated with job retention. They all consider stationary models. Usami (1983) proposes a finite-horizon model with moral hazard, where the probability of reemployment conditional on search depends on the previous *employment* history. Usami confirms the aforementioned decreasing benefits result, and finds that the worker compensation should be non-decreasing during the employment period. Although our model permits employment history dependence, most of our findings are induced by skill depreciation during *unemployment*. Moreover, Usami studies the problem choosing an “inconvenient” state variable, which prevents a complete analysis. Our recursive formulation results in a manageable value function, which allows us to characterize in detail the optimal scheme, both qualitatively and quantitatively.

Literature on human capital depreciation, job displacement and duration dependence We have already mentioned above that Keane and Wolpin (1997) find important wage depreciation rates during unemployment. In recent years, a great deal of attention has also been paid to the consequences of *worker displacement*. Displacement is usually defined as the involuntary separation of workers from their jobs without cause (i.e. for economic reasons) and without future recall. Research on the effects of worker displacement has grown dramatically in recent years, especially in the in the United States (for surveys, see Hamermesh, 1989; Faber, 1993, 1997; Hall, 1995; Fallick, 1996; and Kletzer, 1998). Using a variety of methods and data sets, the findings are remarkably consistent. Displaced workers face large and persistent earnings losses upon reemployment in the order of 10 – 25% compared with continuously employed workers (Jacobson et al., 1993; Ruhm, 1987; and Bartel and Borjas, 1981).⁵ Evidence for European labor markets is contrasting and not always comparable. For example, recently Burda and Mertens (2001) use self-reported information on job-displacement and estimate an average wage growth reduction of approximately 3.6%, with a peak of 17% (for high pay jobs) for Germany. Lefranc (2002) use micro data from labour force surveys and find wage losses upon displacement in the order of 10 – 15%. In contrast, Bender et al. (1999) use information on plant closing to identify displacement, and estimated near zero losses both in France and Germany.⁶ A number of labor economists interpret wage depreciation as reflecting the destruction of firm-specific or industry-specific human capital associated with tenure. Other

⁵In addition to earning losses, displaced workers experience more unemployment than non displaced workers (see - for example - Hall, 1995; Ruhm, 1991; Swain and Podgursky, 1991).

⁶Leonard and van Audenrode (1995) find results similar to the ones of Bender et al. for Belgium, and so does Ackum (1991) for Sweden.

authors pointed out that non-observable individual heterogeneity may bias estimated tenures upwards, so that previous tenure might have a positive effect on post-displacement wage rates (see, for example Kletzer, 1989). Another interpretation is simply the destruction of rents associated with good matches, with no return to tenure per se (Mincer and Jovanovic, 1981; Altonji and Shakotko, 1987; Abraham and Faber, 1987; Ruhm, 1990; Altonji and Williams, 1992).

One of the distinctive features of many current European labor markets is the high proportion of workers that remain unemployed for a long period of time. This feature of the European labor market is widely regarded as a serious problem and has attracted a lot of attention both for efficiency and equity reasons. The *negative duration dependence* in the exit rate from unemployment is felt as one of the major causes for long-term unemployment. In the literature, labor economists suggest different justifications for the observed negative duration dependence in the exit probability from unemployment. A first stream of research supports a purely statistical explanation for this phenomenon. It is argued that the hazard rate estimations are affected by an unobservable heterogeneity effect, which induces a fictitious negative duration dependence in the estimated hazard rates. The idea is that aggregate hazard rates decrease since the sample characteristics are changing through unemployment. In particular, the pool of long-term unemployed, is dominated by the share of people whose individual hazard rates are lower than the average.

However, a number of studies document that even after controlling for unobservable heterogeneity a non negligible negative duration dependence still remains. For the US, van den Berg and van Ours (1996) use CPS data and find that white male exit probability “genuinely” decreases by 30% after 3 months of unemployment. In Europe, van den Berg and van Ours (1999) find that for French young workers the exit probability decreases by 30 – 35% after 2.5 years of unemployment. For the UK, Nickel (1979) find a 50% decrease after 60 weeks of unemployment. More recently, van den Berg and van Ours (1994) find that the British male exit rate of unemployment decreases by 20% after one quarter and by more than 30% after 6 months of unemployment (for the UK see also Lynch, 1985; for Spain see Bover et al., 2002).⁷ A second stream of research attempts to give a theoretical explanation for the mentioned “true” negative duration of the hazard rate. It is indeed empirically documented that during unemployment there is a negative time dependence in the arrival rate of job-opportunities.

⁷In other European countries the result are again somehow contrasting. For a summary on a number of OECD countries, see Machin and Manning (1999).

Heckman and Borjas (1980) call this phenomenon “occurrence dependence.” The analysis of the genuine duration dependence can be divided in two categories. On one hand, we have what we might call supply side explanations. A long-term unemployed worker finds it more difficult to know the existence of vacancies, either because the worker loses valuable social contacts,⁸ or because the long-term unemployed worker suffers some sort of stigmatization by the other workers in the supply side of the market (Gregg and Wadsworth, 1996). The stock-flow approach to search of Coles and Smith (1994) and Gregg and Petrongolo (1997) gives another supply side explanation for the negative duration dependence phenomenon. The idea is that the observed occurrence dependence is in fact only apparent. What really happens is that, at the beginning of unemployment the worker faces and evaluates a stock of vacancies while, in latter periods of unemployment, only the newly open vacancies are processed. Finally, a third motivation is that the hazard rate decreases during unemployment simply because the worker reduces his search effort level. This may be due to a sort of discouraged worker effect that may induce long term unemployed to remain in the labor market, but actually looking almost passively for new jobs,⁹ or because skills and work habits atrophy during unemployment (Sinfield, 1981). We then have a few demand side - or firm-hiring behavior - explanations for the negative duration dependence. Recently, a lot of attention has been devoted to study the so called “stigma effect”. The idea behind this approach is the following. It is assumed that firms imperfectly test workers prior to hiring them. If some firms hire only workers who pass the test, then there is an informational externality; unemployment duration is a signal of workers’ productivity and firms tend to avoid to even test for hiring long-term unemployed workers (Vishwanath, 1989; Lockwood, 1991; Belzil, 1995; and Yoshiaki, 1997). Finally, Blanchard and Diamond (1994) propose a “ranking” model, where it is simply assumed that, independently from any payoff-relevant motivation, firms hire the workers with the lowest unemployment duration levels.

Outline of the paper In the next section, we first present the general formulation of the model. We then summarize the main characteristics of the optimal scheme under stationarity (in Section 2.1). In Section 2.2, we introduce human capital depreciation in a closed form example, and provide sufficient conditions for the existence of endogenous lower bounds on worker’s lifetime utility. In Section 3, we describe our approach to solve the problem, and

⁸See, for example, Calvó-Armengol (2003) for a theoretical analysis of this aspects of the labor market.

⁹Think for example of a worker who is learning about his ability to find a new job. A long period of unemployment will obviously imply a downward biased perception of his hazard rate.

provide some qualitative characteristics of the optimal contract. In Section 4, we calibrate the model both with the US and Spanish economies, and simulate the efficient schemes. Section 5 concludes.

2 Model

Consider a risk-neutral planner who must design an optimal unemployment compensation scheme for a risk-averse worker. The worker has time invariant preferences of the following separable form:

$$u(c) - v(a)$$

where c is consumption and a is the search effort. We assume $u(\cdot)$ to be strictly increasing, strictly concave and continuously differentiable, with bounded inverse function u^{-1} .¹⁰ In any period, the worker can be either employed (e) or unemployed (u). If the worker is employed, he produces the quantity $S(h)$ (his gross wage) which is assumed to be a continuous, increasing, and bounded function of the worker's human capital endowment h . Moreover, we assume that h follows the following stochastic law of motion:

$$h' = m_z(h), \quad z = u, e; \quad \text{with } m_u(h) \leq h \leq m_e(h); \quad \text{and } m_z(\cdot) \text{ continuous,} \quad (1)$$

where h' is the next period human capital level and z is the worker's employment state. The idea is that during unemployment ($z = u$) h depreciates, while during employment there can be human capital accumulation due, for example, to on-the-job training.

While unemployed, the worker can either search ($a = 1$) or non-search ($a = 0$) for a new job, i.e. $a \in A = \{0, 1\}$. The search activity is costly: $v(1) = v > v(0) = 0$, and a affects the transition probability between employment states, according to a hazard rate function $\pi(a, h)$, with $\pi(1, h) \equiv \pi(h) \geq 0$ increasing with h . When the worker does not search, the job finding probability is zero, i.e. $\pi(0, h) \equiv 0$. The situation where the planner requires the agent to stop searching for a job can be interpreted as that of "social assistance" or "early retirement, and it is considered as an absorbing one."¹¹ In order to have a simple tractable recursive formulation,

¹⁰This latter assumption is merely a technical one. It allow us to simplify considerably the proof of Proposition 3.

¹¹In the numerical exercise, we allow for a non absorbing social assistance state with $\pi(0, h) > 0$.

we call “human capital” the aggregate variable h , which affects both the worker’s productivity and his re-employment probabilities. However, h can well be multidimensional, capturing - for example - different degrees of human capital specificity. In fact, the nature of h should be considered more broadly than mere skill or ability. Notice that since we do not study remedies for skill depreciation, such as training programs, our analysis is not affected by the specific nature of h .¹²

The timing of the model during unemployment is as follows. At the beginning of period t the worker receives the unemployment benefit b_t and is required to supply the costly job-search effort a_t . If at the beginning of following period the worker is employed he receives a net wage $w_{t+1} = S(h_{t+1}) - \tau_{t+1}$ otherwise he receives the UI benefit transfer b_{t+1} , and so on.

The crucial assumption of the model that we keep throughout the paper is that the planner cannot observe the worker’s search effort a . Thus, during unemployment there is a moral-hazard problem. This means that unemployment benefits are not paid only as insurance, but must also play the role of giving incentives for search. We assume there are no informational problems related with h .¹³

Following the recursive contracts literature, we characterize the contract using the following recursive formulation. Let U and h be the discounted utility promised to the agent in period t , and its human capital endowment respectively. Given (U, h) , the planner’s value function in the unemployment state V is defined as follows:

$$V(U, h) = \max_{a \in \{0,1\}} \{V_a(U, h)\}. \quad (2)$$

The function V_1 describes the planner’s value in the case where the agent is required to actively

¹²For the purposes of this paper, the model proposed here is consistent with most theoretical explanations of wage depreciation and duration dependence offered in the introduction. A relevant case of non stationarity which is excluded from our analysis is the situation where the worker learns about his labor market opportunities. However, notice that our specification of the model is for example consistent with the “stigma effect.” In this case, the learning activity is done by the firms hence, once the stigma effect is appropriately taken into account in the definition of $\pi(h)$, the worker’s problem does not depend on the particular (demand side) reasons of the hazard rate duration dependence.

¹³Note that, since the laws m_z are known, and the realized employment states z are perfectly observable, it suffices to assume that the planner knows the initial endowment h_0 .

search for a job, and solves

$$V_1(U, h) = \sup_{b, U^u, U^e} -b + \beta [\pi(h)W(U^e, h') + (1 - \pi(h))V(U^u, h')] \quad (3)$$

s.t. :

$$U = u(b) - v + \beta [(1 - \pi(h))U^u + \pi(h)U^e], \quad (4)$$

$$U \geq u(b) + \beta U^u, \text{ and} \quad (5)$$

$$h' = m_u(h).$$

Equation (4) requires the contract to deliver the promised level of discounted utility to the worker, and is called the promise keeping constraint and plays the role of law of motion for the state variable U . Constraint (5) is the incentive compatibility constraint ensuring the agent is willing to deliver the amount of effort called for in the contract.

When the worker is not required to search ($a = 0$), the planner's value is

$$V_0(U, h) = V_0(U) = \max_{b, U^u} -b + \beta V_0(U^u) \quad (6)$$

s.t. :

$$U = u(b) + \beta U^u.$$

In the “assistance” state the planner pays the worker a constant transfer equal to $b = u^{-1}((1 - \beta)U)$ forever. Hence $V_0(U) = -\frac{u^{-1}((1-\beta)U)}{1-\beta}$.

The function $W(U, h)$ denotes the planner's net return in the employment state when the worker is required to receive a lifetime utility level of U , and is endowed with human capital stock h . During this state, the planner satisfies the promise keeping restriction

$$U = u(w) - l + \beta U^e \quad (7)$$

by transferring a net wage w , after imposing the tax (or paying the subsidy) $\tau \equiv S(h) - w$ on the gross wage $S(h)$. We denote by $l \geq 0$ the effort cost of working. In the model jobs are permanent, and we assume that there are no informational asymmetries during employment. While employed, the worker is hence fully insured, and from (7) one can easily verify that the planner's value is

$$W(U, h) = \frac{S_e(h) - w(U)}{1 - \beta} = \frac{S_e(h) - u^{-1}((1 - \beta)U + l)}{1 - \beta}, \quad (8)$$

where $S_e(h)$ is the average discounted gross wage, and can be computed recursively as follows

$$S_e(h) = (1 - \beta)S(h) + \beta S_e(m_e(h)).$$

The properties of u and S , guarantee that W is bounded, strictly decreasing, strictly concave and continuously differentiable in U .

2.1 The Stationary Benchmark

In the stationary version of the model, the re-employment probability $\pi(h) \equiv \pi > 0$ and the gross wage $S(h) \equiv S > 0$ do not depend on the level of human capital h . Similarly to the general case, the value of unemployment V is defined as

$$V(U) = \max_{a \in \{0,1\}} \{V_a(U)\} = \max \left\{ V_1(U), -\frac{u^{-1}((1-\beta)U)}{1-\beta} \right\},$$

where V_1 is the stationary analogous of (3)-(5), and $W(U) = \frac{S - u^{-1}((1-\beta)U)}{1-\beta}$.

Proposition 1 *Assume that at U_0 in period $t = 0$ the planner decides to implement $a_0^* = 1$. Then (i) the agent will never be required to stop searching actively for new jobs, i.e. $a_t^* = 1$ for all t ; (ii) both U_t , b_t and w_t are strictly decreasing during unemployment; and (iii) for any level of utility $\underline{U} > \frac{u(0)}{1-\beta}$, there exists a finite unemployment spell duration T , such that $U_T < \underline{U}$.*

All proofs are reported in the appendix.

Since u is concave, when U_0 is very large even compensating the agent for the search effort cost becomes too expensive, and the planner implements $a_0^* = 0$. For more moderate levels of initial lifetime utility, the worker is required to search intensively for a new job ($a_0^* = 1$). In this case, because of dynamic incentive provision, both the worker's UI payments and lifetime utility decrease during unemployment, and lower lifetime utilities imply lower incentive and effort compensation costs. As a consequence, in a non trivial stationary problem there is no role for social assistance: the worker is always required to search for a job, and the UI transfers never stop decreasing. In addition, because of consumption smoothing, (ii) implies that wage tax $\tau_t = S - w_t$ upon re-employment increases with unemployment duration. The increasing wage tax result is a key finding in the analysis of Hopenhayn and Nicolini (1997a).

The fact, that in order to spread out incentive costs, the planner reduces the agent's expected discounted utility U_t through time is a property always true in stationary models, and have sometimes unpleasant consequences. Point (iii) shows that the optimal contract implied by the repeated moral hazard model creates a weaker form of the “immiserization result”: when u is unbounded below, efficiency requires that worker's expected discounted utility falls, with positive probability, below any arbitrary negative level. The infinite punishments result is questionable in some circumstances. For example, it may be impossible for the planner to enforce, ex-post, such punitive plans because these would imply excessive social conflict costs. Similarly, excessive punishments may induce the worker to opt out of the insurance scheme. Below we show that when human capital depreciates rapidly enough misery is avoided since the optimal scheme generates an endogenous lower bound on payments, hence on lifetime utilities.

In Figure 1, we report a parametrized example of the closed form of $V(U)$ derived in Pavoni (2002) for the case where the worker has logarithmic utility. The value function $V(U)$ (the solid line in the figure) is represented by the upper envelope of the two functions V_0 and V_1 (the dotted lines in the figure). The flatter line represents $V_0(U)$: the planner's return when the worker is fully insured. The steeper line corresponds to $V_1(U)$, the case where the planner always induces the worker search intensively for a new job. The crossing point between the two curves is where the planner “switches” between the two regimes. For future reference notice, that at the switching point the function $V(U)$ is neither concave nor differentiable. If U_0 lies to the left of the switching point and U_t decreases with t then $a_t^* = 1$ for all t as we have just shown.

2.2 Emergence of Endogenous Lower Bounds on Lifetime Utility: A Closed Form Example

We now use a closed form example to show a sufficient condition on the speed of human capital depreciation that guarantee the emergence of a lower bound on worker's lifetime utility. Consider the following relationship between gross wage and human capital endowment:¹⁴

$$S(h) = \omega h > 0.$$

¹⁴This formulation can be motivated by an aggregate technology in which skill units h are perfect substitute. In this case, ω is the skill's marginal product.

Closed Form. Clearly, the *stationary* case can now be replicated by assuming that h remains constant (i.e., $m_u(h) = h = m_e(h)$). In this case, when the utility of the agent takes the logarithmic form, i.e. $u(c) = \ln(c)$, it can be shown that the following solutions to the functional equations (2)-(8) describe the “true” planner value functions:¹⁵

$$V_1(U) = \frac{\omega h K}{1 - \beta} - \frac{B \exp \{(1 - \beta)U\}}{1 - \beta} \text{ if } U \leq M, \quad (9)$$

$$V_0(U) = -\frac{\exp \{(1 - \beta)U\}}{1 - \beta} \text{ if } U \geq M; \text{ and} \quad (10)$$

$$W(U) = \frac{\omega h}{1 - \beta} - \frac{\exp \{l\} \exp \{(1 - \beta)U\}}{1 - \beta} \text{ for all } U. \quad (11)$$

where $M = \frac{\ln(\frac{K\omega h}{B-1})}{1-\beta}$, $K = \frac{\beta\pi}{1-(1-\pi)\beta}$, and B solves

$$\ln \left[B^{\frac{1}{\beta}} - (1 - \pi)B \right] = \ln \pi + \frac{1 - \beta}{\beta} \frac{v}{\pi} + l. \quad (12)$$

It can be verified directly by computing the optimality conditions that when during unemployment $a = 1$, the worker’s lifetime utility decreases according to:¹⁶

$$U^u(U) = U - \frac{\ln B}{\beta}.$$

For later use, notice that - since $\beta < 1$ - equation (12) implies that $\frac{1-\beta}{\beta} \frac{v}{\pi} + l$ constitutes a lower bound for the decrease in worker’s lifetime utility $U - U^u(U)$.

Wage Depreciation. Now assume that during unemployment h depreciates according

¹⁵The reader can verify directly that the proposed functions are a solution of the functional equations. In Pavoni (2002), I show that they actually describe the true value for the planner.

¹⁶From (12) it can be easily shown that $B > 1$. We also have

$$\begin{aligned} U^e(U) &= U + \frac{\frac{v}{\pi} - \ln B}{\beta}, \text{ and} \\ u(b(U)) &= (1 - \beta)U + \ln B. \end{aligned}$$

Moreover, the employment state policy is simply a constant utility for each period, guaranteed by a within-period net wage equal to

$$u(w(U)) = (1 - \beta)U + l.$$

to

$$h_{t+1} = m_u(h_t) = (1 - \delta) h_t,$$

with $\delta > 0$. For simplicity, also assume that during employment h_t remains constant (i.e., $m_e(h) \equiv h$).

Let a_t^* the optimal search effort implemented in period t . It is easy to see that if $a_t^* = 1$ for all t (as it would be the case for the stationary model), the planner would otherwise insure the agent against gross wage depreciation and the ratio between $V_1(U_t, h_t)$ and $V_0(U_t)$ would satisfy

$$\frac{V_1(U_t, h_t)}{V_0(U_t)} \leq \frac{\omega h_0 K \left(\frac{1-\delta}{1-\gamma} \right)^t - B \exp \{ (1-\beta) U_0 \}}{-\exp \{ (1-\beta) U_0 \}} \quad (13)$$

where $1 - \gamma = \exp \left\{ - (1 - \beta) \left[\frac{1-\beta}{\beta} \frac{v}{\pi} + l \right] \right\}$. Notice that when $(1 - \gamma) > (1 - \delta)$, as $t \rightarrow \infty$ the right hand side of (13) tends to $-B < -1$. Hence, there must exist a $T < \infty$ such that $V_0(U_T) \geq V_1(U_T, h_T)$, and the conjecture that $a_t^* = 1$ for all t is false. The program must hence have an endogenous lower bound on U since, when $a_t^* = 0$, U_t remains constant.

Proposition 2 *Consider the log-utility specification of the model, with $\pi(h) = \pi$, $S(h) = \omega h$, $m_e(h) = h$, and $m_u(h) = (1 - \delta) h$. If $(1 - \delta) \leq \exp \left\{ - (1 - \beta) \left[\frac{1-\beta}{\beta} \frac{v}{\pi} + l \right] \right\}$, the scheme generates a lower bound on lifetime utilities and on payments.*

If we normalize $S(h_0) = 100$, a reasonable value for the search effort and work costs v and l would be for example $v = l = 1$.¹⁷ Moreover, the value of 0.08 constitutes a lower bound for the US weakly hazard rate π .¹⁸ Easy numerical computations, after setting $\beta = 0.999$, suggest that a weakly human capital depreciation rate of 0.1% would imply the emergence of a lower bound on U . Notice, that according to Keane and Wolpin (1997), the US weakly value of δ for white males varies between 0.2% and 0.8%.

¹⁷Since $\ln S(h_0) = \ln 100 = 4.6$, an effort cost of 1 is between one fourth and one fifth of this value.

¹⁸According to Meyer (1990), for example, $\pi = 0.1$.

3 Qualitative Analysis

We now study our model in its general specification. It is well known that hidden-action moral-hazard models do not typically describe concave problems (Grossman and Hart, 1983, and Phelan and Townsend, 1991). There are four main reasons why this may prove to be problematic, especially in a dynamic environment. First of all, non concavity might also lead to the non differentiability of the problem. Second, even assuming differentiability, first-order conditions may not longer be sufficient for local maxima. Third, the analysis might become more complicate since we are actually looking for a global maximum. Finally, and more importantly, the usual envelope theorems¹⁹ cannot be applied, and this may reduce considerably the usefulness of our recursive formulation.

In this section we develop a systematic approach that allows for these complications. First, we somehow confirm the above mentioned difficulties. We find that “in most cases” the associated value function is neither concave nor differentiable. We however find a positive result as well: the optimal contract can still be characterized to a great extent by using the familiar first order conditions.

The idea of our approach is as follows. The complication involved by the recursive study of the dynamic moral-hazard problem comes from the incentive constraint. This prevents a direct approach to the study of the concavity and differentiability of the value function V . We thus first reformulate the problem to make it suitable for such analysis. We define a collection of concave and continuously differentiable functions (*the conditional functions*), of which the value function V is the upper envelope. We then apply the extended envelope theorem of Daskin (1967) to this problem to show that V is almost everywhere differentiable.²⁰

Our successive step is to study the “switching points,” that is, the utility levels at which the upper envelope function V switches between two different conditional functions of the above mentioned class. Those points are indeed the only problematic ones. However, given the characteristics of our class of functions, each switching point possesses a very nice characteristic: either the V function is in fact differentiable at this point, or the point is never reached in equilibrium. The fact that the points of non-differentiability cannot be reached in

¹⁹For envelope theorems we refer to theorems that describe conditions under which the value function of a parametrized optimization problem is a differentiable function of parameter.

²⁰In the economic literature, this result was rediscovered by Kim (1993), Sah and Zhao (1998), Milgrom (1999), and recently extended by Milgrom and Segal (2002). None of these papers mention that the Daskin’s result can be applied to solve the dynamic moral hazard problem. We also show that in dynamic models the characteristics of the value function permits the use of the first order conditions.

equilibrium allows us to disregard them while characterizing the optimal contract.

3.1 The sequence of efforts formulation and the existence result

Consider the space $\mathcal{A}$ of all the sequences of efforts $\mathbf{a} = \{a(n)\}_{n=0}^{\infty}$ $a(n) \in A$, implementable during unemployment. For any human capital endowment $h \in \mathcal{H}$, effort sequence $\mathbf{a} \in \mathcal{A}$ and utility level $U \in \mathcal{U} \subset \mathbb{R}$ we can define

$$\begin{aligned} V(\mathbf{a}, U, h) &= \sup_{b, U^e, U^u} -b + \beta [\pi(a, h)W(U^e, h') + (1 - \pi(a, h))V({}_1\mathbf{a}, U^u, h')] \quad (14) \\ &\text{s.t. (1), and} \\ &\text{if } a = 1, \text{ (4) and (5); if } a = 0, U = u(b) + \beta U^u. \end{aligned}$$

The function(al) $V(\mathbf{a}, U, h)$ represents the planner's optimal payoffs *conditional on* a given sequence of efforts, when the worker is unemployed. The symbol ${}_1\mathbf{a} = \{a(n)\}_{n=1}^{\infty}$ stands for the “one-step-ahead” continuation of $\mathbf{a}$.

It would be easy to show that the value function of the sequential problem satisfies (14).²¹ In the next Proposition we show that also the converse is true.

Proposition 3 *The Bellman operator implied by (14) defines a contraction in the space of continuous and bounded functions with the sup norm. Thus V exists and is unique, and*

$$\|V\|_{\infty} = \sup_{y \in Y = \mathcal{A} \times \mathcal{U} \times \mathcal{H}} |V(y)| < \infty.$$

In the next Proposition, we show that each conditional function $V(\mathbf{a}, U, h)$ has nice properties, very similar to the ones of the employment state value function W .

Proposition 4 *Consider a sequence of efforts $\mathbf{a} \in \mathcal{A}$ and an endowment level h , together with a law m . Let $V(\mathbf{a}, U_0, h) = -b_0 + \beta [\pi(a_0, h)W(U_0^e, h') + (1 - \pi(a_0, h))V({}_1\mathbf{a}, U_0^u, h')]$ with U_0 in the interior of the effective domain of $V(\mathbf{a}, \cdot, h)$, and with b_0 belongs to the interior of the domain of the agent's utility function u . The conditional function $V(\mathbf{a}, \cdot, h)$ is concave and continuously differentiable in U at any such U_0 , and*

$$V'(\mathbf{a}, U_0, h) \equiv \frac{\partial V(\mathbf{a}, U_0, h)}{\partial U} = -\frac{1}{u'(b_0)} < 0 \quad (15)$$

²¹The technical reader should be reassured by the fact that we do not have problems of measurability, since we have only two outcomes.

The problem (14) of implementing optimally (minimizing costs) a *given* sequence of efforts $\mathbf{a}$ is concave, with linear constraints. The associate value function $V(\mathbf{a}, U, h)$ is thus concave as well. Given concavity, differentiability can be shown by applying the Benveniste and Scheinkman (1979) Lemma in the standard way.

We now show that the maximization with respect to $\mathbf{a}$ is always well defined. Hence, the value function $V(U, h)$ defined can be written as the upper envelope of the collections of conditional functions $V(\mathbf{a}, U, h)$.

Proposition 5 *The set $\mathcal{A}$ of sequences of efforts is compact and $V(\mathbf{a}, U, h)$ is continuous in $\mathcal{A}$ for any (U, h) . Thus, a maximum exists for any (U, h) , and we can define*

$$V(U, h) = \max_{\mathbf{a} \in \mathcal{A}} V(\mathbf{a}, U, h). \quad (16)$$

3.2 The shape of the value function

Proposition 5 defines $V(U, h)$ as the upper envelope of the collection of the conditional functions $V(\mathbf{a}, U, h)$. The approach we propose exploits this interpretation for $V(U, h)$. In order to eliminate the possibility of weird behaviors at the infinite due to the non-stationarity of the problem, we first make the following regularity assumption:

Assumption A1 For any endowment h , there is a time horizon $T(h) < \infty$ such that $\forall t \geq T(h)$ both $S(m^t(0, h)) = \underline{S}$, and $\pi(m^t(0, h)) = \underline{\pi}$.

Assumption **A1** can be seen as a restriction on the law m , or on the functions $S(\cdot)$ and $\pi(\cdot)$, or on both. It should be noted that, the above conditions allow $T(h)$ to be arbitrarily large (provided that it remains finite).

Proposition 6 *Consider a pair (U, h) , with U as in Proposition 4 and assume that **A1** is satisfied. Then $V(U, h)$ possesses both right and left derivative in the first argument, with $V_+(U, h) \geq V_-(U, h)$. Moreover, $V(\cdot, h)$ is almost everywhere differentiable for any h , and where it is differentiable we have*

$$V'(U, h) = V'(\mathbf{a}, U, h) \text{ for all } \mathbf{a} \in \mathcal{A}^*(U, h), \quad (17)$$

where $\mathcal{A}^*(U, h)$ is the (non-empty) set of efforts solving the maximization (16) defined in Proposition 5.

Proposition 6 is based on the Daskin's envelope theorem. In order to apply this result to our problem, we need to guarantee that the derivative $V'(\mathbf{a}, U, h) \equiv \frac{\partial V(\mathbf{a}, U, h)}{\partial U}$ is jointly continuous in $(\mathbf{a}, U)$. Our line of proof is simple. Notice that any finite periods version of our model satisfies the joint continuity requirement of the partial derivative $V'(\mathbf{a}, U, h)$. The reason is that in a finite time horizon problem the set of all possible paths of efforts $\mathcal{A}$ is finite. From Proposition 4 we know that $V'(\mathbf{a}, U, h)$ is continuous in U alone, and the finiteness of $\mathcal{A}$ implies joint continuity in $(\mathbf{a}, U)$ as well. Under Assumption **A1** the infinite horizon version of the model can be reduced to the case with a finite set of effort sequences.

Finally notice, that from (15) and (17), at the points where V is differentiable, one recovers the usual envelope condition. This property will be used below.

3.3 Characterization of the optimal contract

Clearly, the characteristic $V'_+(U, h) \geq V'_-(U, h)$ we derived above is not a property of the concave functions. In fact, when the directional derivatives differ, $V(U, h)$ cannot be concave in any interval containing U . This confirms the analysis of the stationary model (see Figure 1 and the closed form (9)-(10)). However, notice that the “kink” at the switching point in Figure 1 has a particular nature: it is an “inward” one. It turns out that this characteristic of V is also true in the general formulation of the model and implies that the optimal choice of the continuation utility U^u will never be at a switching point. The problem (3) is hence differentiable at all relevant points, and the usual first-order conditions, although obviously non sufficient, are still a necessary characteristic of an efficient contract. This is our main result in Proposition 7.

Proposition 7 *Assume **A1** and interiority. The optimal contract necessarily satisfies*

$$V'(U, h) = -\frac{1}{u'(b^*)} \quad (18)$$

$$W'(U^{e*}, h') = -\frac{1}{u'(b^*)} - \mu \quad \mu \geq 0 \quad (19)$$

$$V'(U^{u*}, h') = -\frac{1}{u'(b^*)} + \frac{\pi(a^*, h)}{1 - \pi(a^*, h)} \mu, \quad (20)$$

with $\mu = 0$ if either $a^* = 0$ or (5) is satisfied with strict inequality. Moreover, (18) can possibly fail only in the first period. In addition, we have

$$V'(U, h) = [\pi(a^*, h)W'(U^{e*}, h) + (1 - \pi(a^*, h))V'(U^{u*}, h)].$$

The implications for the optimal scheme are not yet transparent. By rearranging the above derived first-order conditions and using envelope theorem we get the following:²²

Corollary 8 *Under the conditions of the previous proposition we have the following*

$$\frac{1}{u'(b_t^*)} = \pi(a_t^*, h_t) \frac{1}{u'(w_{t+1}^*)} + (1 - \pi(a_t^*, h_t)) \frac{1}{u'(b_{t+1}^*)}. \quad (21)$$

Moreover, (i) $w_{t+1}^* \geq b_t^* \geq b_{t+1}^*$; and (ii) either is true that $w_{t+1}^* > b_t^* > b_{t+1}^*$, or $w_{t+1}^* = b_t^* = b_{t+1}^*$.

Result (i) confirms, in a general, possibly non stationary framework, one key finding of Shavell and Weiss (1979) and Hopenhayn and Nicolini (1997) for the stationary model: the optimal unemployment insurance scheme requires the UI benefits to decrease with the duration of unemployment. However notice, that the wage tax behavior remains indeterminate. In Section 4, we take advantage of the recursive formulation to perform computer simulations of the optimal contract for the US and the Spanish economy. We anticipate that in our numerical exercises we find that an optimal scheme typically generates a plan of wage taxes $\tau_t^* = S_t - w_t^*$ which decreases with the length of previous unemployment spell. This is of course in contrast with the result of the stationary model of Section 2.1, where the optimal re-employment wage tax τ_t^* increases during unemployment.

The second part of Corollary 8 has another important implication. In many studies, unemployment insurance programs are modelled in a very simple way. Only two parameters are used to define the scheme. It is assumed to be a time invariant unemployment benefit payment b , which - usually because of job-search incentives - is strictly lower than a time

²²Although equation (21) in Corollary 8 basically coincides with that derived - using a variational approach - by Rogerson (1985), our first order conditions (18)-(20) provide a stronger characterization of the optimal contract. This becomes particularly evident in a dynamic moral hazard model with more than two outcomes. In this case, equation (21) alone typically would not permit to say much about the monotonicity of the payments. In contrast, our first order conditions (which we easily conjecture to be true), together with the envelope theorem, would for example allow us to link punishments and rewards to the likelihoods ratio, confirming in a multiperiod environment the results of the static model.

invariant wage payment w . We can then ask the following question. Could this simple scheme be optimal, for some combination of wage depreciation and duration dependence? Part (ii) gives a clear negative answer to this question.

In exchange, we provide an alternative, simple way of describing an optimal scheme, or at least some of its key characteristics. To make our results ready to use for policy purposes, we rearrange (21) and obtain

$$\frac{1}{u'(b_t^*)} - \frac{1}{u'(b_{t+1}^*)} = \pi(a_t^*, h_t) \left[\frac{1}{u'(w_{t+1}^*)} - \frac{1}{u'(b_{t+1}^*)} \right],$$

which, for the case with logarithmic utility, reduces to

$$b_t^* - b_{t+1}^* = \pi(a_t^*, h_t) [w_{t+1}^* - b_{t+1}^*]. \quad (22)$$

Because of its graphical representation, shown in Figure 2, we may name the above equation as *the triangle rule*. According to (22), for small $\pi(a_t^*, h_t)$, optimality suggests almost flat UI schemes with a relatively large difference between net wage and unemployment insurance benefits. And vice versa for high hazard rates. This gives to (22) a quite appealing economic interpretation. According to this condition, workers facing relatively low hazard rates should be motivated to search for new jobs mainly through rewards: in case they find a new job, they should receive a high net wage w_{t+1}^* . In contrast, to those workers facing high probabilities of reemployment, search incentives are mainly given by the use of punishments: in case of failure in the job-search process, on those workers the planner imposes a considerable drop in the unemployment benefit payment b_{t+1}^* .

The triangle rule can be used as a very simple, “back of the envelope”, test of optimality for a unemployment insurance scheme. For that consider an existing scheme. For each period t , using today and tomorrow’s benefit payment levels and tomorrow’s net wage, it is always possible to draw a triangle such as that in Figure 2. The test also needs a reliable point estimate of the hazard rate $\pi(a_t^*, h_t)$ associated with unemployment duration of length t . Then using $\pi(a_t^*, h_t)$ one can easily check whether the two parts that form the segment $w_{t+1}^* - b_{t+1}^*$ have the proportions required by equation (22). When this is not the case, the trade-off between insurance and incentives is not exploited optimally, and there is room for a budget saving reform.

Alternatively, one could consider condition (22) as a valuable tool for deriving an optimal unemployment insurance scheme. If the policy maker had full knowledge of the hazard rate

functions $\pi(\cdot, \cdot)$, he could design the unemployment scheme using the triangle rule as a minimal restriction for efficiency.

Finally, note that the triangle rule might not be satisfied if we introduce other constraints into the designing problem, such as the exogenous lower bound on worker's utility I imposed in Pavoni (2003). Thus - assuming an optimal government behavior - the triangle rule could also be used to test for the existence of exogenous minimum bounds; which is a way to discriminate between the unrestricted model of Hopenhayn and Nicolini (1997a,b) and the restricted one with utility bounds studied in Pavoni (2003).

A further quantitative assessment would regard the degree of approximation - toward the fully optimal scheme - that the use of the triangle rule would involve for utility functions different from the logarithmic one. Although the log-utility case seems to be a good approximation for an average level of relative risk aversion,²³ this analysis is left for future research.

4 Quantitative Analysis

More than giving a detailed policy advice, the aim of this section is to illustrate the effect of both wage depreciation and hazard rate duration dependence on the shape of the efficient UI scheme. In particular, our aim is to disentangle the effects on the optimal program of each one of these two consequences of human capital depreciation.

Wage depreciation and duration dependence clearly have the effects of reducing the planner's returns from the search activity and increasing the incentive costs for any *given* level of worker's lifetime utility. However, the analysis of the stationary model shows that because of dynamic incentive provision, the worker's expected discounted utility is decreasing during unemployment, and low lifetime utilities imply lower incentive and effort compensation costs. Obviously, which one of these two forces - human capital depreciation or decrease in lifetime utility - dominates in a non stationary model is a quantitative issue.

We hence pursue our task by presenting a couple of simple calibration exercises. Our first example refers to Spain. Spanish data on wages are often poor or difficult to interpret. We thus use the Spanish economy to study how the optimal unemployment insurance scheme is

²³Attanasio and Weber (1993) use UK cohort data to estimate the intertemporal elasticity of substitution. Assuming CRRA preferences, the results of Attanasio and Weber imply a constant risk aversion parameter between 1.3 and 1.5 (where 1 corresponds to the log-case). This is consistent with many other previous studies. For example, Mehra and Prescott (1985) cite various empirical studies that provide support for a constant relative risk aversion parameter between 1 and 2.

affected by duration dependence in the probability finding a new job *alone*. In our second example, we use the US economy to study the implications of wage depreciation.

Our numerical methodology is based on value function iteration. We approximate the value function with Chebyshev polynomials.²⁴ Value function iteration involves a maximization step. Although first-order conditions were successfully used to characterize qualitatively the contract, given the non-concavity of the problem, we are forced to use a numerical global maximization procedure to determine the optimal choices at each value function iteration.

In both exercises we assume log-utility preferences over consumption, i.e. $u(c) = \ln c$, and we set the search cost equal to the cost of labor, i.e. $v = l > 0$.

4.1 The Spanish Example: the effects of the hazard rate duration dependence.

To calibrate the model with the Spanish economy we assume a *constant* gross wage S . Hence the calibration exercise consists of choosing 3 parameters: $[S, \beta, l]$, together with the hazard rate paths $\{\pi(a, h_t)\}_{t=0}^T$ for $a \in \{0, 1\}$ and the initial utility level U_0 . We normalize the wage S to 100, and interpret each period as a month by setting $\beta = 0.996$. Finally, we set $v = l = 1$, which is between one fourth and one fifth of the utility the agent would receive from consuming the gross wage $u(S) = \ln(100) = 4.6$. In summary $[S, \beta, v = l] = [100, 0.996, 1]$.

The initial level of worker's utility U_0 is computed backward in accordance with the existing scheme. The current insurance system can be represented by a contract that has no duration dependent taxes or transfers when employed ($w = S$), and pays a first benefit level b^1 of 70 for the first six months of unemployment, from the 7th to the 24th month the benefit level b^2 is set equal to 60, and from the 25th onward we assume the worker receives an assistance level of benefits $b^3 = 20$.²⁵ The corresponding expected discounted utility value U_0 for an unemployed

²⁴Chebyshev polynomials have several mathematical and practical advantages as discussed in Judd (1998).

²⁵In Spain, the replacement ratio is equal to 70% during the first six months of unemployment and 60% thereafter, subject of a floor of 75% of the minimum wage. Benefit duration is one-third of the last job's tenure, with a maximum of two years. The assistance system pays, for up to two years, 75% of the minimum wage to (unemployed) workers, with dependant, whose average family income is precisely below that amount. In 1998, the minimum wage was around 70,000 pesetas (\$280) (Guia Laboral 1998 y de Asuntos Sociales (1998)). The amount of the non-contributive assistance level of transfers varies across different Autonomous Communities between 30,000 to 45,000 pesetas (\$150/180), is means-tested, and there is no fixed duration (See Lopez (1996)). The Bulletin of Labor Statistics (1999) reports as 300,000 pesetas (\$1,200) per month the 1998 average wage in non-agricultural activities. Following the common assumption that workers subject to severe unemployment risk face a wage that is two thirds of the average national wage, we consider the assistance level of benefits as $\frac{1}{5} = 20\%$ of the gross wage S .

worker, can be calculated backward as follows. According to the existing Spanish UI scheme, when a worker finds a job, his lifetime utility is

$$U_{work} = \frac{u(S) - l}{(1 - \beta)} = \frac{4.6 - 1}{0.004} = 900,$$

which represents the utility of working forever and receiving the gross wage $S = 100$. Moreover, note that from period T onward the worker's problem is stationary. Both the unemployment benefits and the probability of finding a job are at their minimum level. Thus, under our parametrization, the worker will be searching for a job ($a = 1$), which will be found with probability $\pi(1, h_T) = \underline{\pi}$. The value of his expected discounted utility U_T at period T can be computed as follows

$$U_T = \frac{u(b^3) - 1 + \beta \underline{\pi} U_{work}}{1 - \beta(1 - \underline{\pi})},$$

where $b^3 = 20$ is the non-contributive assistance level of unemployment benefits. For any $0 \leq t \leq T$ we can now define the value U_{T-t} recursively by

$$U_{T-t} = u(b_t) - v(a_t^*) + \beta [\pi(a_t^*, h_t) U_{work} + (1 - \pi(a_t^*, h_t)) U_{T-(t-1)}],$$

where the period t benefit level $b_t \in \{b^1, b^2, b^3\}$ is computed according to the three steps scheme described above, and a_t^* denotes the effort level chosen optimally by the worker in period t .²⁶

Our calibration of the hazard rates is based on Bover, Arellano and Bentolila (2002). According to our interpretation of the data, an increase in unemployment duration seems to reduce the effectiveness of the job-search activity. However, since the search effort level a is obviously not reported by the study we had to perform an “identification” exercise. Here is our strategy. Bover et al. (2002) use data from the Spanish Labour Force Survey (EPA) to estimate the Spanish hazard rates both for workers receiving unemployment insurance benefits and for those who do not receive any UI benefit transfer.²⁷ This for 30 consecutive months

²⁶That is, a_t^* solves

$$\max_{a_t \in \{0,1\}} u(b_t) - v(a_t) + \beta [\pi(a_t, h_t) U_{work} + (1 - \pi(a_t, h_t)) U_{T-(t-1)}].$$

²⁷Notice that the pool of workers non receiving benefits include those who received benefits in some previous

of unemployment duration. As expected, for any duration level of unemployment the former hazard rates (the one related to workers receiving UI benefits) are always lower than the latter ones. Moreover, the difference between the two functions decreases considerably with unemployment duration, and approaches zero after 2 years. We believe it is reasonable to assume that workers not receiving benefits supply a higher effort level than workers receiving benefits. We need to assume that this is the case at any level of unemployment duration. These considerations induce us to interpret in our two effort framework the decrease in the difference between the hazard rates of the two groups of workers as a decrease in the effectiveness of the search activity. As a consequence, we calibrate the hazard rate paths $\{\pi(a, h_t)\}_{t=0}^T$ by linearly interpolating the estimations of Bover et al. (2002) as follows. We set $T = 30$, $\pi(1, h_0) = 0.21$ and $\pi(1, h_T) = 0.03$ with a constant decrease in the hazard rate of 0.06 in each period. Furthermore, we set the “passive” hazard rate $\pi(0, h_t) = \hat{\pi} = 0.01$ at a constant level for any unemployment duration t , which is consistent with the approximate stationarity of the estimated lower hazard rate function.

The results of our simulations of the efficient scheme are reported in Figure 4. The interested reader can first of all verify directly that the scheme obeys to the triangle rule. Second, observe that the optimal path for unemployment benefit payments (the dotted lower-level line) is initially decreasing and then becomes completely flat. Recall that the incentive constraint can be written as follows:

$$U^e \geq U^u + \frac{v}{\beta\pi(h)}.$$

When $\pi(h)$ decreases incentive costs increase since, in order to widen the difference between U^e and U^u , the planner is forced to impose additional uncertainty on worker’s consumption. In addition, the expected returns from the search activity $\pi(h)\frac{S}{1-\beta}$ decrease as well. Hence, if human capital depreciates rapidly enough during unemployment, for long durations the planner does not find profitable to induce the agent to search actively for a new job, and the UI payments eventually stop decreasing because the agent is fully insured. This second part of the benefit path (the flat one) is particularly important since induces an endogenous lower bound on worker’s lifetime utility U . Third, notice that the net wage w_t - represented by the upper-level line - is initially roughly constant and then increases with unemployment duration. For durations longer than 24 months, the worker enters into the assistencial program, and the

period.

net wage displays a remarkable drop since there is no more need for incentive provision.

Since S is constant at 1, our simulations suggest that between 15 and 24 months of unemployment duration it is optimal to pay a wage subsidy upon re-employment. The planner finds profitable to induce the agent to search actively for a subsidized job since the wage subsidy cost is considerably lower than that of paying the UI benefits. The optimality of a wage subsidy is one key prediction of the non stationary of our model. In fact, we already mentioned that this policy implication contrasts with that of stationary models à la Hopenhayn and Nicolini.

The intuition for the behavior of the net wage is as follows. In a multiperiod setting, the optimal incentive scheme is shaped by the tension between within and between period consumption smoothing. The planner can improve within period consumption insurance, i.e., reduce the difference between b_t and w_t , by relegating part of the punishment burden to the future. That is why the lifetime utility U_t and UI payments b_t decrease during unemployment. The emergence of a lower bound on worker's lifetime utility shorthands the effective time horizon of the problem forcing the planner to design a scheme biased toward the static component of the incentives. That is, the planner is forced to widen the difference between UI payments and net wage. Clearly, this effect tends to reduce the steepness of the net wage schedule w_t . However, it would also be present in a stationary model with an *exogenous* minimum bound on U . And, as it is shown in Pavoni (2003), this force alone cannot induce the net wage schedule to increase with unemployment duration, i.e. it cannot induce a wage subsidy when S is constant. We discussed above that when $\pi(h)$ decreases the planner is forced to generate a larger wedge between U^e and U^u because of the incentive constraint. Is this additional effect on w_t that leads to the wage subsidy result in this case.

4.2 The US Example: the consequences of wage depreciation.

Our second numerical example is a calibration with the US economy, and focuses on the effect of wage depreciation. We thus assume that the hazard rate does not vary with h : $\pi(1, h) = \pi$ and $\pi(0, h) = \hat{\pi}$ for all h . Similarly to the analysis of Section 2.2, we consider a very simple linear relationship between human capital endowment and gross wage: $S(h_t) = \omega h_t$ and assume a geometric depreciation rule m_u for human capital: $h_{t+1} = (1 - \delta)h_t$. For simplicity, we also assume that when the worker is employed his human capital endowment remains constant, i.e. $m_e(h) = h$ for all h . To complete our calibration exercise we have six parameters to choose: $[v = l, \beta, \pi, \hat{\pi}, \delta, \omega]$, together with the initial states U_0 and h_0 . Notice, that

in our specification the human capital wage depreciation rates coincide, hence we can fix $\omega = 1$ without loss of generality. We also normalize $h_0 = S_0 = 100$, and set $v = l = 1$. To be consistent with the US payment system, we choose one week as reference period, and accordingly calibrate $\beta = 0.999$. Using the results in Meyer (1990) and partially following Hopenhayn and Nicolini (1997a), we set the weakly US hazard rate at $\pi = 0.1$, the “passive” hazard rate at $\hat{\pi} = 0.01$, and we compute the initial level U_0 in a way similar to that of previous example.²⁸ Finally, the depreciation rate parameter δ is calibrated according to the estimation of Keane and Wolpin (1997). Using NLSY data, they (structurally) estimate an annual human capital depreciation rate for white US males of between 9.6% (for blue collars) and 36.5% (for white collars). We set our weekly level of $\delta = 0.005$, which corresponds to an intermediate annual depreciation rate of 23%. In summary, our choice are as follows

$$\begin{bmatrix} S_0 & l = v & \beta & \pi & \hat{\pi} & \delta \\ 100 & 1 & 0.999 & 0.1 & 0.01 & 0.005 \end{bmatrix}.$$

The results of our simulations are reported in Figure 5. Three lines are displayed in this figure as a function of unemployment duration t : the UI benefit payments b_t (represented by the thick dotted lower-level line), the gross wage $S_t = \omega h_t$ (represented by the homogeneously decreasing solid line), and the net wage $w_t = S_t - \tau_t$ (the thin dotted upper-level line). The optimal path for unemployment benefit payments b_t presents qualitatively the same characteristics as the Spanish case: it is initially decreasing and then - approximately after 60 weeks - becomes completely flat. Notice that in this example both π and $\hat{\pi}$ are constant, hence the incentive costs of implementing the high effort level are constant. The planner’s expected returns are however decreasing, since the gross wage $S(h)$ decreases with unemployment duration. The resulting effect is similar to the one of previous example: at some point the planner “releases” the agent from the search duty. The agent is hence fully insured and the UI payments stop decreasing at some level generated endogenously by the efficient contract (assistance).

Notice that the net wage schedule w_t is very flat at the 93% of the original wage level.

²⁸According to Meyer (1990), the average level of UI benefits received in the sample is 66% of the average value of the pre-unemployment wage $S(h_0) = 100$, and lasted - again in average - 34 weeks. To have a finite value for U_0 with log utility, we assumed that after 34 weeks of UI benefit payments the worker continues to receive 0.1% of his gross wage. The latter can be justified - for example - by the existence of assistance programs such as the Temporary Assistance for Needy Families (TANF) program; or by the existence of charity institutions which provide a minimal level of financial support to individuals.

The reason for this phenomenon is the emergence of the assistance state. We mentioned in the Spanish case that the emergence of a lower bound on worker's lifetime utility tends to reduce the steepness of the net wage schedule since it forces the planner to increase the within period gap between b_t and w_t . We also mentioned there that the analysis of the stationary model suggests that this force alone cannot induce the net wage to increase with unemployment duration. The reason is consumption smoothing. Since U_t decreases with t and w_t is (weakly) increasing in U_t , the net wage cannot increase during the unemployment spell. This characteristic is fully confirmed here. As the analysis of Section 2.2 suggests, since the planner tends to otherwise insure the agent against gross wage shocks, when π does not change with h , the optimal contract displays essentially the same characteristics of that derived under stationary. In this case, it turns out that the optimal way of increasing the difference between b_t and w_t keeping w_t non-increasing in t is to maintain the net wage schedule essentially flat during unemployment.

From the figure, it is clear that the flatness of w_t implies a decreasing wage tax $\tau_t = S(h_t) - w_t$; and for unemployed durations between 12 and 60 weeks is actually optimal to pay a wage subsidy upon re-employment. For example, recall that the parameter δ is calibrated such that after $t = 54$ weeks, the gross wage is $S(h_{54}) = 77$. This implies that, after approximately one year of unemployment, the worker should receive a wage subsidy of $(\frac{93}{77} - 1) 100 = 20.8\%$.

Also in this case, when the worker enters the assistencial program, there is no more need for incentive provision, and the net wage schedule w_t displays an important downward jump.

5 Conclusions

In the present paper we extend previous studies on optimal unemployment insurance to incorporate the effects of human capital depreciation and duration dependence in the mechanism-design problem.

Our results partially confirm those obtained with stationary models, namely that benefits should decrease with unemployment duration. However, the introduction of human capital depreciation and duration dependence generates some novel features on the optimal program. First, if human capital depreciates sufficiently rapidly during unemployment, the planner loses the incentives to induce the agent to supply the high search effort level. Consequentially unemployment benefit payments are initially decreasing and eventually become completely flat, since the long-term unemployed worker is fully insured by the planner. This creates

an endogenous lower bound on worker's expected discounted utility, providing an alternative way of eliminating the immiserization result of Thomas and Worrall. Second, we find that the increasing wage tax result of the stationary models à la Hopenhayn and Nicolini is not robust to this extension. Our simulations for the US and Spanish economies both show that although it is optimal to impose a wage tax after reemployment on short-term unemployed workers, the optimal level of wage tax should decrease with the length of worker's previous unemployment spell, eventually becoming a wage subsidy for the long-term unemployed. The optimality of a *wage subsidy* is in fact a key policy implication of our non stationary search model with moral hazard.

Our analysis also has an independent theoretical interest, in that we develop a new approach that allows us to study recursively the properties of the dynamic moral hazard model in a systematic way. We find that the associated value function is in general non-concave and non-differentiable. In spite of these non-smoothness problems, we show that the optimal contract can be characterized by using the usual first order conditions. The technique we developed in this paper uses the Daskin's envelope theorem, and can be easily extended both to a more general class of moral-hazard problems and to other problems with similar characteristics.

One aspect of our characterization of the contract is a simple necessary characteristic of the efficient UI scheme with log-utility agents, which we call it the triangle rule. We believe that the simplicity of the triangle rule deserves an accurate analysis. We plan to study the degree of optimality of an unemployment scheme computed using the triangle rule, when the worker's utility function is different from the logarithmic case, but remains within a range of parameters consistent with microeconomic estimations.

In both examples of Section 4, a worker who stays unemployed for a sufficiently long period is no-longer required to actively searching for a job, and he is fully insured. Notice that the long-term unemployed worker is a net cost for the government, especially when he has no hope of finding a new job in the future. In this case, an alternative active labor market policy may become the only valuable alternative to rescue the worker from his state. In the near future, we plan to extend our model and analyze more composed welfare-to-work policies such as job search assistance and training programs.

Appendix: Proofs

Proof of Proposition 1 (i) If at $t = 0$ $a_0 = 1$ then we have that $V_1(U_0) \geq V_0(U_0)$. For any $n \geq 0$ let us now define the set of conditional functions as follows

$$\begin{aligned} V(U, n+1) &= \max_{b, U^e, U^u} -b + \beta [\pi W(U^e) + (1 - \pi)V(U^u, n)] \\ U &= u(b) - v + \beta [\pi U^e + (1 - \pi)U^u] \\ U &\geq u(b) + \beta U^u, \end{aligned}$$

with $V(U, 0) \equiv V_0(U)$. Notice they are all strictly concave and continuously differentiable for $U > \frac{u(0)}{1-\beta}$. By the contraction mapping, also $V(U, \infty)$ is concave. It is not difficult to see that $V(U, \infty)$ is continuously differentiable.

We now need the following lemma.

Lemma 9 $V'(U, n+1) \leq V'(U, n)$ for any n with strict inequality at least for some n .

Proof of the lemma. We will follow an inductive argument. Let consider first the problem for $n = 0$:

$$\begin{aligned} V(U, 1) &= \max_{z, U^e, U^u} -g(z) + \beta [\pi W(U^e) + (1 - \pi)V_0(U^u)] \\ U &= z - v + \beta [\pi U^e + (1 - \pi)U^u] \\ U &\geq z + \beta U^u. \end{aligned}$$

It is easy to see that incentive compatibility is binding in this case (since $W'(U) = V'_0(U)$, if the incentive compatibility is slack, we have $U^u = U^e$). Hence from the first order conditions we have

$$-g'(U - \beta U^u) = V'(U, 1) > V'_0(U^u) = -g'((1 - \beta)U^u)$$

which implies $U > U^u$. But then $U - \beta U^u > (1 - \beta)U$. From the convexity of g we have that $V'_0(U) = -g'((1 - \beta)U) > V'(U, 1)$.

Now assume that $V'(U, n) \leq V'(U, n-1)$ and solve the problems in the two cases. From

first order conditions we get

$$\begin{aligned}
V'(U_n^u, n-1) &= -g'(U - \beta U_n^u) + \mu_n \frac{\pi}{1-\pi} \\
W'(U_n^e) &= -g'(U - \beta U_n^u) - \mu_n \\
V'(U_{n+1}^u, n) &= -g'(U - \beta U_{n+1}^u) + \mu_{n+1} \frac{\pi}{1-\pi} \\
W'(U_{n+1}^e) &= -g'(U - \beta U_{n+1}^u) - \mu_{n+1}
\end{aligned}$$

Now assume $\mu_{n+1} > \mu_n \geq 0$. Then by the induction argument we must have $U_n^u \geq U_{n+1}^u$ (just assume that $U_{n+1}^u > U_n^u$ and you get a contradiction), and by the incentive compatibility constraint we have $U_n^e \geq U_{n+1}^e$. But then $U - \beta U_n^u \leq U - \beta U_{n+1}^u$ and that $\mu_{n+1} > \mu_n$ implies that $-g'(U - \beta U_{n+1}^u) - \mu_{n+1} < -g'(U - \beta U_n^u) - \mu_n$ which is in contradiction with the fact that by concavity $W'(U_{n+1}^e) \geq W'(U_n^e)$. Hence it must be that $\mu_n \geq \mu_{n+1} \geq 0$. Similarly as before, the induction argument and incentive compatibility imply $U_n^u \leq U_{n+1}^u$ and by envelope $V'(U, n+1) \leq V'(U, n)$. **Q.E.D.**

Clearly, by the above inequalities and the fact that V is differentiable, the limit function $V(U, \infty)$ is such that $V'(U, \infty) \leq V'(U, n)$. Now notice that we have also shown that if $a_0 = 1$ then $U^u \leq U$. As a consequence Lemma 9 implies that if $a_0 = 1$ will never choose $a_t = 0$ in the future and $V_1(U) = V(U, \infty)$.

(ii) Now since V_1 is concave, at any t , from the first order conditions we have that if $U_t > \frac{u(0)}{1-\beta}$ then $U_t > U_t^u = U_{t+1}$, (in Pavoni, 2003; Lemma 1, I show that the incentive constraint must be binding) hence U_t is decreasing. From the envelope condition b_t decreases as well. From the incentive compatibility constraint we have that U_t^e decreases, and from the definition of W $u(w_t) = (1-\beta)U_t^e$, so w_t also decreases.

(iii) If we let $\underline{U} > \frac{u(0)}{1-\beta}$ and $\delta \equiv \min_{\underline{U} \leq U \leq U_0} \{U - U^u(U)\}$ then the minimization is well defined (by the Maximum Theorem), and $\delta > 0$. Hence the duration of unemployment T is any natural number satisfying $T \geq \frac{U_0 - \underline{U}}{\delta}$. **Q.E.D.**

Proof of Proposition 3 First of all we should define a topology on $\mathcal{A}$. In fact, we do more than that. We define the δ -metric on $\mathcal{A}$ as follows

$$d_\delta(\mathbf{a}, \mathbf{a}') = \sum_{n=0}^{\infty} \delta^n |a(n) - a'(n)|, \quad \delta \in (0, 1).$$

Second, let us simplify the notation by eliminating the h indexation. It will become clear below that the continuity of $m_u(\cdot)$, together with Lemma 9.5 of Stokey et al. (1989) (SLP), allows us to make this simplification, at this stage.

Now consider a generic $s = (\mathbf{a}, U)$ with $\mathbf{a} = \{a(0), a(1), a(2), \dots\} = \{a(0), {}_1\mathbf{a}\} \in \mathcal{A}$. The distance between any two points is $\|s - s'\| = \|\mathbf{a} - \mathbf{a}'\|_\delta + |U - U'|$. Using the promise keeping constraint we rewrite the Bellman operator T as follows

$$\begin{aligned} (TV)(s) = & \sup_{U^u, U^e; \text{sub}(5)} -u^{-1}(U - v(a(0)) - \beta[\pi(a(0))U^e + (1 - \pi(a(0)))U^u]) + \\ & + \beta[\pi(a(0))W(U^e) + (1 - \pi(a(0)))V({}_1\mathbf{a}, U^u)] \end{aligned}$$

Now we show that the operator T maps bounded and continuous functions into itself. From the definition of continuity, we must verify that for each given point s and for each $\varepsilon > 0$, there exists a $\gamma > 0$ such that

$$\text{if } \|s - s'\| < \gamma \quad \text{then} \quad |(TV)(s) - (TV)(s')| < \varepsilon.$$

To this extent, we rewrite the previous condition using the definition of the Bellman operator

$$\begin{aligned} & \left| \sup_{U^u, U^e; \text{sub}(5)} g(U, U^u, U^e, a(0)) + \beta(1 - \pi(a(0)))V({}_1\mathbf{a}, U^u) + \right. \\ & \left. - \sup_{U^u, U^e; \text{sub}(5)} g(U', U^u, U^e, a'(0)) + \beta(1 - \pi(a'(0)))V({}_1\mathbf{a}', U^u) \right| < \varepsilon \end{aligned} \quad (23)$$

where

$$g(U, U^u, U^e, a(0)) = -u^{-1}(U - v(a(0)) - \beta[\pi(a(0))U^e + (1 - \pi(a(0)))U^u]) + \beta\pi(a(0))W(U^e)$$

and

$$W(U^e) = \frac{S - u^{-1}((1 - \beta)U^e + l)}{1 - \beta}.$$

Now consider the two subcases.

Case 1: Suppose that $a(0) = a'(0)$. In this case, we can assume $a(0)$ as a parameter of the

problem, and apply the Maximum Theorem to the problem

$$\begin{aligned} G(U, {}_1\mathbf{a}) &= \sup_{U^u, U^e} g(U, U^u, U^e, a(0)) + \beta(1 - \pi(a_0))V({}_1\mathbf{a}, U^u) \\ \text{sub} \quad &: \quad (5), U^u, U^e \in \Gamma(U) \end{aligned} \quad (24)$$

to show continuity of G in $(U, {}_1\mathbf{a})$. The auxiliary constraint $\Gamma(U)$ is imposed in order to guarantee the constraint correspondence to be compact valued. A possibility is the following. The incentive compatibility constraint (5) requires $U^e \geq U^u + f(a(0))$, so we can always choose appropriately two constants $k_1, k_2 > 0$, and add to (5) the constraints $U^u \geq U - k_1$ and $U^e \leq U + k_2$. The continuity of G implies that we can always find a γ such that (23) is verified.

Case 2: Now suppose $a(0) \neq a'(0)$. The idea here is that we do not check for continuity in this case, that is, we set γ such that, whenever $a(0) \neq a'(0)$, then $\|s - s'\| > \gamma$. This can always be done since in this case $|a(0) - a'(0)| = 1$. In summary, the choice of γ is done according to the continuity properties of G , with the restriction $\gamma \leq 1$.

Since u^{-1} is bounded, if we start from a bounded V , TV is bounded as well.

Finally, one can checked directly that the operator satisfies the Blackwell's sufficient conditions, thus T defines a contraction in the complete metric space of the bounded and continuous functions with the sup norm, in the “reduced” space $S = \mathcal{A} \times \mathcal{U}$. The continuity of the low $m_u(\cdot)$, allows us to complete the proof by applying Lemma 9.5 and Theorem 9.6 SLP, which guarantee that the contraction mapping result is still true in the original space $\mathcal{A} \times \mathcal{U} \times \mathcal{H}$, with h as exogenous sate variable. ■

Proof of Proposition 4 The presence of h creates only notational complications, so we fix it and eliminate the h index in what follows. Following Grossman and Hart (1983) and changing the variable by defining $z \equiv u(b)$, the problem becomes

$$\begin{aligned} V(\mathbf{a}, U) &= \sup_{z, U^u, U^e} -u^{-1}(z) + \beta[\pi(a)W(U^e) + (1 - \pi(a))V({}_1\mathbf{a}, U^u)] \\ \text{s.t.} \quad &: \quad z - v(a) + \beta[(1 - \pi(a))U^u + \pi(a)U^e] \geq z - v(\hat{a}) + \beta[(1 - \pi(\hat{a}))U^u + \pi(\hat{a})U^e] \\ U &= z - v(a) + \beta[(1 - \pi(a))U^u + \pi(a)U^e] \end{aligned} \quad (25)$$

where a is the first element in the sequence $\mathbf{a} = \{a(n)\}$. Notice that the problem satisfies all the conditions required to apply Theorems 4.7 and 4.8 of SLP. To see why the prob-

lem is monotone, use the promise keeping constraint and notice that since u is increasing, the planner's objective function $-u^{-1}(U - v(a) + \beta[(1 - \pi(a))U^u + \pi(a)U^e])$ is strictly decreasing in U . In particular, notice that interiority is important here, it guarantees that $z = U - v(a) + \beta[(1 - \pi(a))U^u + \pi(a)U^e]$ can indeed be modified to satisfy promise keeping without affecting incentive compatibility. Finally, note that if V is concave the planner objective function is concave (since u^{-1} is convex), and the constraints set is convex (linear), as a consequence (25) is a concave problem. This proves concavity.

Differentiability can be shown as follows. Given the value function is concave we can use Lemma 2 in Benveniste and Scheinkman (1979). For a fixed level of promised utility U_0 we are looking for a differentiable concave function $G(\mathbf{a}, U)$ such that it is well defined in an interval I around U_0 and such that for any $U \in I$ we have $G(\mathbf{a}, U) \leq V(\mathbf{a}, U)$ and $G(\mathbf{a}, U_0) = V(\mathbf{a}, U_0)$. We claim that

$$G(\mathbf{a}, U) = -u^{-1}(U + v(a) - \beta[(1 - \pi(a))U_0^u + \pi(a)U_0^e]) + \beta[pW(U_0^e) + (1 - p)V(\mathbf{a}, U_0^u)]$$

is the function we are looking for. Indeed, the optimal values U_0^e and U_0^u satisfy the incentive compatibility and (by interiority) the promise keeping constraint can always be satisfied by varying the benefit transfer b , so the function G is well defined and we have $G(\mathbf{a}, U) \leq V(\mathbf{a}, U) \forall U \in I$ as required. The properties of u imply the concavity and differentiability of $-u^{-1}$. So G is concave and differentiable, and this implies that V is differentiable in U_0 and $V'(\mathbf{a}, U_0) = -\frac{1}{u'(b_0)}$. Since V is concave, it is continuously differentiable. ■

Proof of Proposition 5 Given the continuity result we obtained in Proposition 3, to show Proposition 5 it suffices to show the compactness of $\mathcal{A}$. That is

Lemma 10 $\mathcal{A}$ is compact in topology induced by the metric $d_\delta(x, y)$ for any $\delta \in (0, 1)$.

Proof Noting that the set of all infinite sequences of zeros and ones corresponds to the Cantor set $\Delta \equiv \{0, 1\}^{\mathbb{N}}$, which is known to be compact in the topology induced by the metric $d_\delta(x, y)$ for $\delta = \frac{1}{3}$.²⁹ From this we can easily show that the Cantor set is topologically equivalent to the same set endowed with the topology induced by a generic $\delta \in (0, 1)$. Finally notice that the set $\mathcal{A}$ is a closed set of such set of sequences, hence it is compact. **Q.E.D.**

²⁹If we start by endowing the set $\{0, 1\}$ with the discrete topology (which is both compact and Hausdorff), then it is well known that the metric $d_\delta(x, y) = \sum_{n=0}^{\infty} \delta^n |x(n) - y(n)|$ with $\delta = \frac{1}{3}$ induces the product topology on Δ (see, for example, Aliprantis and Border 1994, page 93) so from the properties of the Hausdorff spaces and using the Tychonoff Product Theorem we have that Δ is both Hausdorff and Compact.

Given that existence is proved, it is then easy to combine the continuity and boundedness results of Proposition 5 to show the equivalence between the sequential and the recursive choice of efforts. ■

Proof of Proposition 6 Our line of proof is based on the Daskin's envelope theorem. To state the result in terms of our model we first need some definitions.

Definition 1 For each U and h , define the non-empty set $\mathcal{A}^*(U, h) = \arg \max_{\mathbf{a} \in \mathcal{A}} V(\mathbf{a}, U, h)$, moreover we call $\mathcal{A}^*(h) = \bigcup_U \mathcal{A}^*(U, h)$ the set of all possible maximizers.

For completeness, we now state the Daskin's envelope theorem and provide a version of the proof adapted from Milgrom (1999).

Lemma 11 Assume that (i) $\mathcal{A}^*(h)$ is non-empty and compact and that for any U, h , the function $V(\cdot, U, h) : \mathcal{A}^*(h) \rightarrow \mathbb{R}$ is continuous. Moreover assume that (ii) $V'(\mathbf{a}, U, h) \equiv \frac{\partial V(\mathbf{a}, U, h)}{\partial U}$ exists and is continuous in $(\mathbf{a}, U)$. Then, for any h , the value function $V(U, h)$ has always both right and left derivative in U , and these are given by the formulas

$$\begin{aligned} V'_+(U, h) &= \max_{\mathbf{a} \in \mathcal{A}^*(U, h)} V'(\mathbf{a}, U, h) \\ V'_-(U, h) &= \min_{\mathbf{a} \in \mathcal{A}^*(U, h)} V'(\mathbf{a}, U, h), \end{aligned}$$

moreover $V(U, h)$ is almost everywhere differentiable in U , and whenever the derivative exists then

$$V'(U, h) = V'(\mathbf{a}^*, U, h) \quad \text{for any } \mathbf{a}^* \in \mathcal{A}^*(U, h).$$

Proof of the lemma. To simplify the notation we neglect the dependence on h . Hence $\mathcal{A}^*(U, h)$ becomes $\mathcal{A}^*(U)$, and $\mathcal{A}^*(h)$ becomes $\mathcal{A}^*$.

Let us show first the right hand derivative. From our assumptions $\mathcal{A}^*(U)$ is non-empty and u.h.c., so for each U we can take $\mathbf{a}(U) \in \arg \max_{\mathbf{a} \in \mathcal{A}^*(U)} V(\mathbf{a}, U)$. This is indeed a well defined procedure since $V(\mathbf{a}, U)$ is continuous. Now consider $U' > U$ and write the incremental ratio

$$\frac{V(U') - V(U)}{U' - U} = \frac{V(\mathbf{a}(U'), U') - V(\mathbf{a}(U), U)}{U' - U} \geq \frac{V(\mathbf{a}(U), U') - V(\mathbf{a}(U), U)}{U' - U}$$

the last inequality comes from the fact that $\mathbf{a}(U') \in \mathcal{A}^*(U')$ so any other choice will reduce the value of $V(\mathbf{a}(U'), U') = V(U')$. Now from the properties of the *conditional* functions as

$U' \rightarrow U$ with $U' > U$ the right hand side converges to the derivative. So we have that

$$\liminf_{\substack{U' \rightarrow U \\ U' > U}} \frac{V(U') - V(U)}{U' - U} \geq V'(\mathbf{a}(U), U).$$

And we are done for the first part. We now want to show that

$$\limsup_{\substack{U' \rightarrow U \\ U' > U}} \frac{V(U') - V(U)}{U' - U} \leq V'(\mathbf{a}(U), U)$$

where I recall that $\mathbf{a}(U)$ is the maximizer of the partial derivative $V'(\mathbf{a}, U)$ over the non-empty and compact set $\mathcal{A}^*(U)$. To this extent, notice that we have the following

$$\frac{V(U') - V(U)}{U' - U} \leq \frac{V(\mathbf{a}(U'), U') - V(\mathbf{a}(U'), U)}{U' - U} = V'(\mathbf{a}(U'), \xi). \quad (26)$$

The first inequality in (26) is guaranteed by a similar reasoning as the one we made before. Namely, because $\mathbf{a}(U')$ does not in general maximize V when the utility is at U instead of at U' . The last equality, i.e. the existence of a point $\xi \in (U', U)$ such that it is true is guaranteed by the Mean Value Theorem. Now, since ξ and the value of the derivative is calculated for a given $\mathbf{a}(U')$, which in turn depends on U' , we write this number $\xi(U')$. From the continuity of $V'(\mathbf{a}, U)$ we know that for any U' , the partial derivative in the right hand side $V'(\mathbf{a}(U'), \xi(U'))$ is a real number, hence the number

$$R = \limsup_{\substack{U' \rightarrow U \\ U' > U}} V'(\mathbf{a}(U'), \xi(U'))$$

is well defined. Since R is an accumulation point, there exists a converging subsequence $R_n \rightarrow R$. Moreover, we take a sequence of U_n such that this sequence is reproduced by points of the type $V'(\mathbf{a}(U_n), \xi(U_n))$. The reason is that $V'(\mathbf{a}, U)$ is continuous jointly in $(\mathbf{a}, U)$. Since R_n converges, there must exist a converging sequence $(\mathbf{a}(U_n), \xi(U_n)) \rightarrow (\bar{\mathbf{a}}, U)$. The sequence of $\xi(U_n)$ must converge to U since for any n $\xi(U_n) \in (U_n, U)$ and $U_n \rightarrow U$. Moreover, by construction, the sequence of $\mathbf{a}(U_n)$ is such that for any n , $\mathbf{a}(U_n) \in \mathcal{A}^*(U_n)$. If the correspondence $\mathcal{A}^*(U)$ were upper-hemicontinuous, then we could be sure that $\bar{\mathbf{a}} \in \mathcal{A}^*(U)$. This is nothing more than the definition of upper-hemicontinuity: an upper-hemicontinuous correspondence is defined by the fact that $\mathbf{a}(U_n) \in \mathcal{A}^*(U_n)$ for any n implies that $\bar{\mathbf{a}} = \lim_{n \rightarrow \infty} \mathbf{a}(U_n)$ is such that $\bar{\mathbf{a}} \in \mathcal{A}^*(\lim_{n \rightarrow \infty} U_n) = \mathcal{A}^*(U)$. But notice that since the partial derivative $V'(\mathbf{a}, U)$ is

continuous in U , $V(\mathbf{a}, U)$ is continuous also in U (not necessary jointly in $(\mathbf{a}, U)$). So the Theorem of the Maximum guarantees that $\mathcal{A}^*(U) = \arg \max_{\mathbf{a} \in K} V(\mathbf{a}, U)$ is upper-hemicontinuous correspondence. We then have

$$\limsup_{\substack{U' \rightarrow U \\ U' > U}} \frac{V(U') - V(U)}{U' - U} \leq \limsup_{\substack{U' \rightarrow U \\ U' > U}} V'(\mathbf{a}(U'), \xi(U')) = V'(\bar{\mathbf{a}}, U) \leq V'(\mathbf{a}(U), U)$$

where the first inequality comes from the first part of equation (26) by taking the sup on both sides. The second equality comes from our assumptions and the last inequality holds because $\bar{\mathbf{a}} \in \mathcal{A}^*(U)$ and $\mathbf{a}(U)$ is the maximizer of $V'(\mathbf{a}, U)$ in this set. ■

Proof of Proposition 6 Notice that after period $T(h)$ the problem becomes stationary. Hence from Proposition 1 we know that the optimal path of actions $\mathbf{a}^*$ is necessarily such that either (I) for all $t \geq T(h)$ we have $a(t) = 0$ or (II) $a(t) = 1$ for all $t \geq T(h)$. In both cases the set of optimal efforts $\mathcal{A}^*(h)$ for the period-zero problem, is a subset of the *finite* set of all the sequences of efforts which end either with the sequence $\mathbf{0} = \{0, 0, 0, 0, \dots\}$ or with the sequence $\mathbf{a} = \mathbf{1}$ after $T(h) < \infty$.

From Proposition 5 the set $\mathcal{A}^*(h)$ is non-empty. Moreover, it is also trivially compact, since it is finite. This guarantees that assumption (i) of Lemma 11 is satisfied. Finally, notice that from Proposition 4 $V'(\mathbf{a}, U, h) \equiv \frac{\partial V(\mathbf{a}, U, h)}{\partial U}$ exists and it is continuous in U . Since $\mathcal{A}^*(U, h)$ is finite (and non-empty), $V'(\mathbf{a}, U, h)$ is also jointly continuous in $(\mathbf{a}, U)$. Hence also assumption (ii) is satisfied, and we can apply Lemma 11 to our problem. ■

Proof of Proposition 7 It is immediate to see that (18), (19) and (20) are the first order conditions for the proposed problem. Moreover, notice that the existence of $V'(U, h)$ is justified by Proposition 4. However, we must show: (i) first, that the differentiability conditions for taking the first order conditions are indeed satisfied, (ii) second, that $\mu \geq 0$ as claimed.

We will use the following lemma.

Lemma 12 *Assume that f is a continuous function that admits both right and left derivatives in an interior point U_0 . If we have $f'_-(U_0) < f'_+(U_0)$ then U_0 cannot be a maximum.*

Proof. Consider the incremental ratio for the left derivative $f'_-(U_0)$. If $\frac{f(U) - f(U_0)}{U - U_0} < 0$ for $U - U_0 < 0$, with U sufficiently closed to U_0 , we must have $f(U) > f(U_0)$, which is a contradiction to U_0 being a maximum. A similar argument can be used for the right derivative.

Q.E.D.

(i) Since the case with $a^* = 0$ is obvious, we will consider only $a^* = 1$. When $a^* = 1$, the incentive constraint (5) can be rewritten as follows

$$U^e - U^u \geq \frac{v}{\beta\pi(h)}. \quad (27)$$

We can have two cases. *Case 1:* At the optimum the incentive constraint (27) is satisfied with equality. If we rewrite the objective function using (27) with equality and use (4), we can rewrite the problem as a function of U^u alone:

$$\sup_{U^u} -u^{-1}(U - \beta U^u) + \beta \left[\pi(h)W\left(U^u + \frac{v}{\beta\pi(h)}\right) + (1 - \pi(h))V(U^u, h') \right].$$

The problem is now a free maximization whose objective function is a weighted sum between the differentiable functions u^{-1} and W , and the function $V(U^u, h')$. We can directly apply Lemma 12 to this problem and obtain the desired result. *Case 2:* The optimum is such that the incentive constraint (27) is slack. In this case, we can use (4) and rewrite the problem as a function of both U^u and U^e as follows

$$\sup_{U^u, U^e} -u^{-1}(U + v - \beta[U^u + \pi(h)(U^e - U^u)]) + \beta[\pi(h)W(U^e, h) + (1 - \pi(h))V(U^u, h')].$$

Notice that in the objective function the two choice variables U^u and U^e interact in a very peculiar way. Either they are part of a linear mapping into a differentiable function (this is the case if the first term of the objective function, inside u^{-1}) or they enter into two different function which are related linearly among them. This feature guarantees that when taking the directional derivative for optimality we can separate the two variables. The choice of U^e is clearly well defined since both u^{-1} and W are differentiable everywhere. Moreover, for any given choice of U^e , the optimal level U^u is now computed by solving again a *free* maximization over a weighted sum between the differentiable function u^{-1} and the function $V(U^u, h')$, thus Lemma 12 also applies to this case.

(ii) Consider again the $a^* = 1$ case. Once we have shown that the problem must be differentiable at the optimum, we can use the (local) Kuhn-Tucker theorem. For this, notice that the incentive constraint (27) is linear, hence satisfies the constraint qualification requirement needed to apply the Kuhn-Tucker theorem. Hence, if μ is the multiplier associated to the incentive constraint, μ is non-negative as claimed. ■

Proof of Corollary 8 The first part of the corollary is easily derived from the last result of Proposition 7. It suffice to use Proposition (4) and rewrite $V'(U, h) = -\frac{1}{u'(b_t^*)}$, $W'(U^{e*}, h') = -\frac{1}{u'(w_{t+1}^*)}$, and $V'(U^{u*}, h') = -\frac{1}{u'(b_{t+1}^*)}$. To show the second part, notice that since $\pi(1, h) > 0$, both (i) and (ii) results can be easily derived from (18), (19), (20) $\mu \geq 0$ and the strict concavity of u . Obviously, if $a_t^* = 0$ then $w_{t+n}^* = b_t^* = b_{t+n}^*$ for $n \geq 1$. ■

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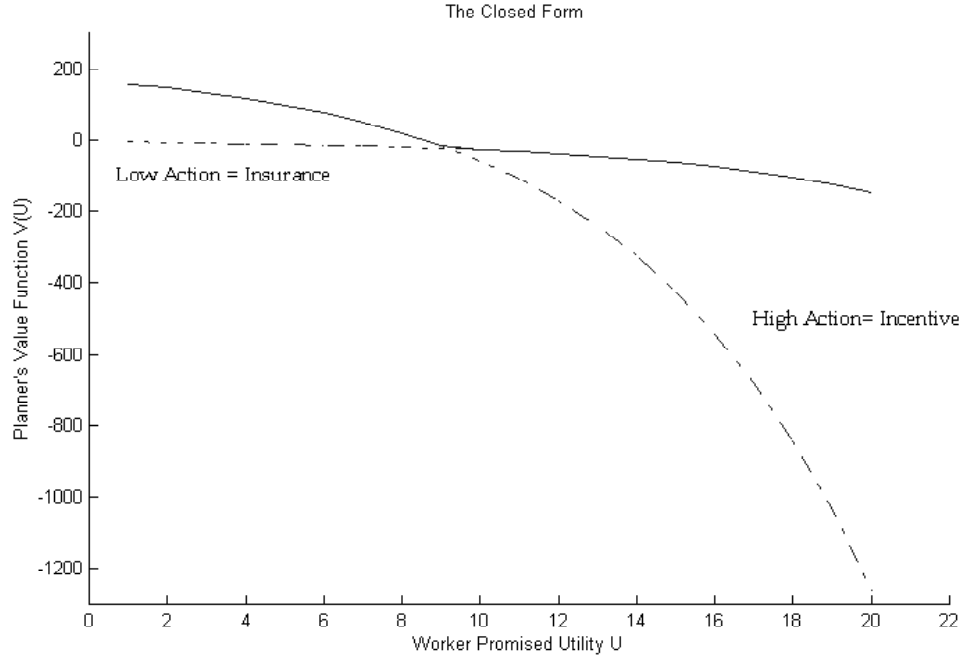


Figure 1: A parametrized example of the Closed form ($u = \ln$). The unemployment state value function $V(U)$ is represented by the upper envelope of the two functions. The flatter dotted line represents $V_0(U)$, the planner's return in case the worker is fully insured. The steeper dotted line corresponds to $V_1(U)$, the case where in any period the planner induces the worker to search actively for a new job. From the picture it is clear that at the switching point the value function $V(U)$ is neither concave nor differentiable.

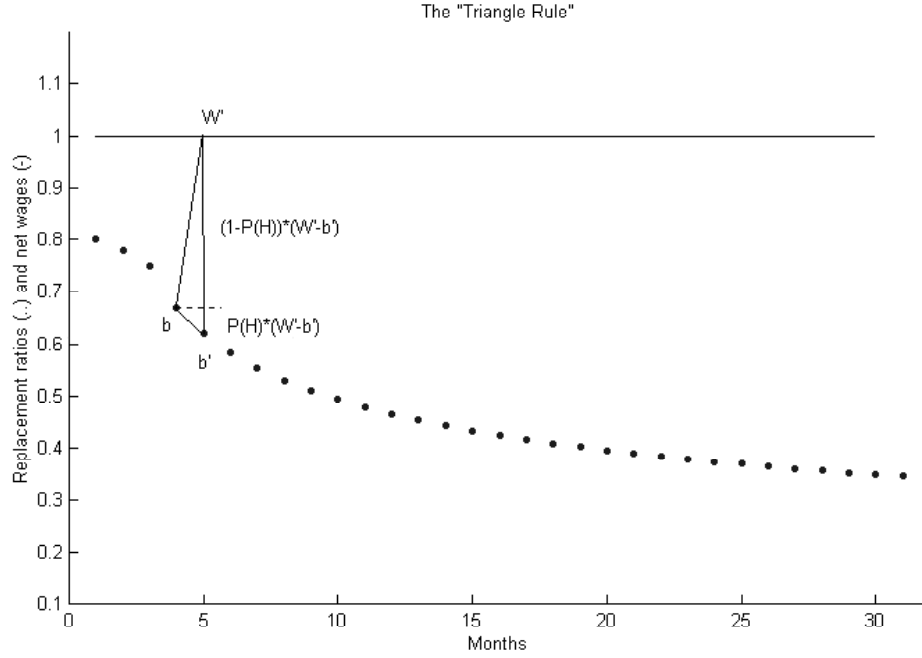


Figure 2: An example of the triangle rule. Given a UI scheme, using today and tomorrow's benefit payments and tomorrow's net wage, it is always possible to draw a triangle such as that in the figure for any duration t . Given an estimate of the hazard rate $\pi(a_t^*, h_t)$ one can easily check whether the two parts that form the segment $w_{t+1}^* - b_{t+1}^*$ have the required proportions.

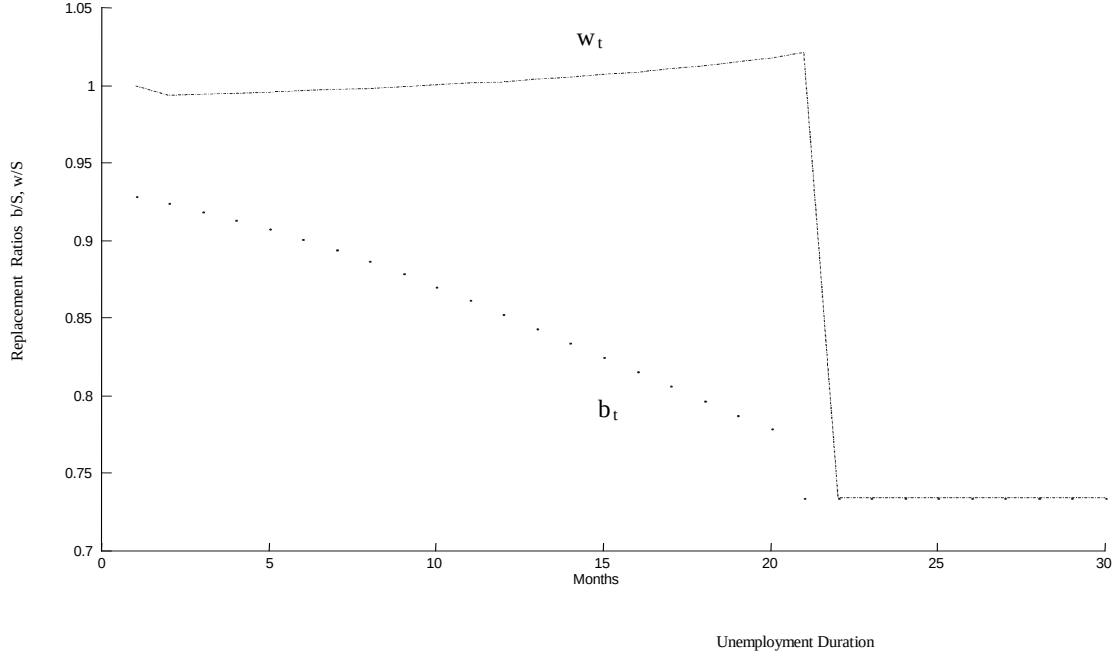


Figure 3: The Spanish Example: the effects of the hazard rate duration dependence. In this figure, the dotted lower-level line represents UI benefit payments b_t , and the upper-level line represents net wage $w_t = S - \tau_t$, as a function of unemployment duration. From the figure, b_t decreases in t and given that the gross wage S is constant at 1, in our example, τ_t is initially decreasing in t , and it then presents a last downward jump. Between the 15th and the 22th month $w_t > S$, i.e., the planner pays to the worker a wage subsidy.

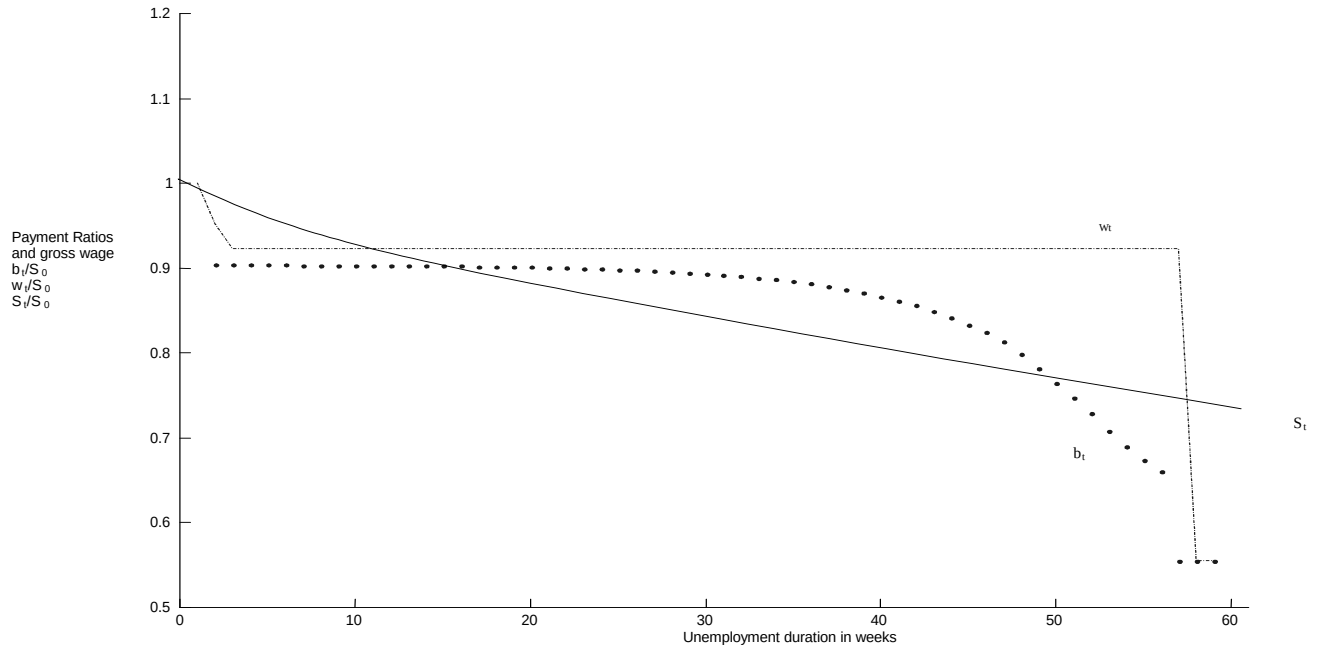


Figure 4: **The US Example: the consequences of wage depreciation.** In this figure, the thick dotted lower-level line represents UI benefit payments b_t , the homogeneously decreasing solid line represents the gross wage $S_t = S(h_t)$, and the thin dotted upper-level line represents net wage $w_t = S_t - \tau_t$, as a function of unemployment duration. The figure shows that b_t is decreasing in t and the reemployment wage tax τ_t steadily decreases until the 58th week, where it is negative, i.e. the planner transfer becomes a substantial wage subsidy.